

PROJECT BASED EXPERIMENTAL LEARNING

COURSE- DATABASE MANAGEMENT SYSTEM (NoSQL / MongoDB)

PROJECT REPORT

TOPIC- MONGODB STUDENT DATABASE OPERATIONS



SUBMITTED TO- MR. Harendra Singh Rajpoot

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3rd semester

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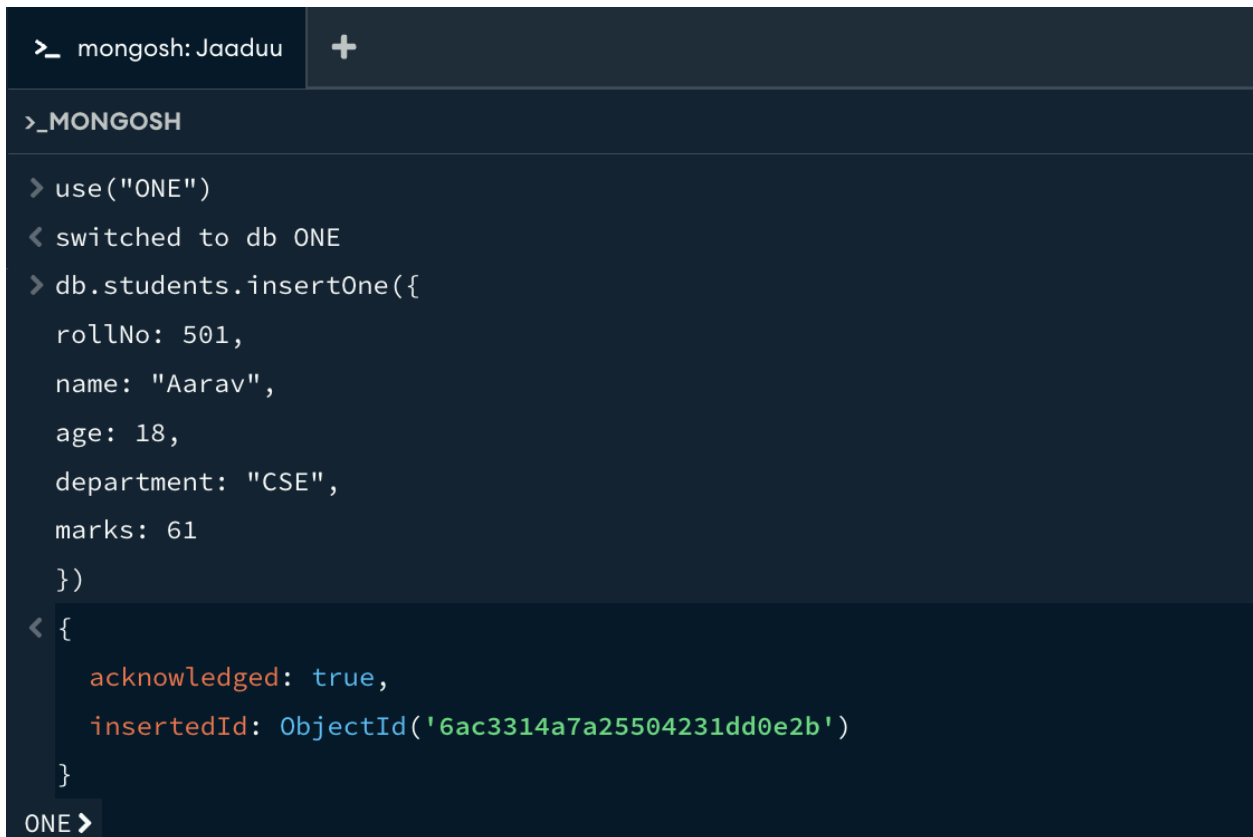
Lucknow, U.P

1. Insert One

Inserts a single student document into the students collection.

Code:

```
db.students.insertOne({
  rollNo: 501,
  name: "Aarav",
  age: 18,
  department: "CSE",
  marks: 61
})
```



The screenshot shows a MongoDB terminal window with a dark theme. The title bar at the top reads '>_ mongosh: Jaaduu' followed by a '+' icon. The main area shows the following commands and output:

```
>_MONGOSH
> use("ONE")
< switched to db ONE
> db.students.insertOne({
  rollNo: 501,
  name: "Aarav",
  age: 18,
  department: "CSE",
  marks: 61
})
< {
  acknowledged: true,
  insertedId: ObjectId('6ac3314a7a25504231dd0e2b')
}
```

At the bottom left, there is a prompt 'ONE >'.

2. Insert Many

Inserts multiple student documents (roll numbers 502 to 551) into the collection at the same time.

Code:

```
db.students.insertMany([
  { rollNo: 502, name: "Vivaan", age: 20, department: "IT", marks: 75 },
  { rollNo: 503, name: "Aditya", age: 19, department: "ECE", marks: 68 },
  { rollNo: 504, name: "Kabir", age: 18, department: "ME", marks: 94 },
  { rollNo: 505, name: "Ishaan", age: 18, department: "CSE", marks: 97 },
  { rollNo: 506, name: "Reyansh", age: 21, department: "IT", marks: 62 },
  { rollNo: 507, name: "Ayaan", age: 18, department: "ECE", marks: 65 },
```

{ rollNo: 508, name: "Vihaan", age: 19, department: "ME", marks: 74 },
{ rollNo: 509, name: "Arnav", age: 22, department: "CSE", marks: 98 },
{ rollNo: 510, name: "Krishna", age: 18, department: "IT", marks: 95 },
{ rollNo: 511, name: "Dhruv", age: 19, department: "ECE", marks: 94 },
{ rollNo: 512, name: "Yuvraj", age: 21, department: "ME", marks: 74 },
{ rollNo: 513, name: "Shaurya", age: 21, department: "CSE", marks: 97 },
{ rollNo: 514, name: "Rudra", age: 20, department: "IT", marks: 60 },
{ rollNo: 515, name: "Advait", age: 19, department: "ECE", marks: 87 },
{ rollNo: 516, name: "Kiara", age: 20, department: "ME", marks: 77 },
{ rollNo: 517, name: "Anaya", age: 19, department: "CSE", marks: 73 },
{ rollNo: 518, name: "Diya", age: 20, department: "IT", marks: 66 },
{ rollNo: 519, name: "Myra", age: 18, department: "ECE", marks: 84 },
{ rollNo: 520, name: "Aadhya", age: 18, department: "ME", marks: 82 },
{ rollNo: 521, name: "Saanvi", age: 20, department: "CSE", marks: 98 },
{ rollNo: 522, name: "Aarohi", age: 20, department: "IT", marks: 62 },
{ rollNo: 523, name: "Ira", age: 21, department: "ECE", marks: 94 },
{ rollNo: 524, name: "Navya", age: 18, department: "ME", marks: 84 },
{ rollNo: 525, name: "Prisha", age: 18, department: "CSE", marks: 95 },
{ rollNo: 526, name: "Riya", age: 20, department: "IT", marks: 99 },
{ rollNo: 527, name: "Siya", age: 20, department: "ECE", marks: 96 },
{ rollNo: 528, name: "Anvi", age: 19, department: "ME", marks: 64 },
{ rollNo: 529, name: "Zara", age: 18, department: "CSE", marks: 74 },
{ rollNo: 530, name: "Pihu", age: 20, department: "IT", marks: 65 },
{ rollNo: 531, name: "Om", age: 19, department: "ECE", marks: 66 },
{ rollNo: 532, name: "Rehan", age: 21, department: "ME", marks: 77 },
{ rollNo: 533, name: "Veer", age: 21, department: "CSE", marks: 83 },
{ rollNo: 534, name: "Aryan", age: 19, department: "IT", marks: 83 },
{ rollNo: 535, name: "Dev", age: 20, department: "ECE", marks: 73 },
{ rollNo: 536, name: "Ansh", age: 20, department: "ME", marks: 64 },
{ rollNo: 537, name: "Yash", age: 22, department: "CSE", marks: 70 },
{ rollNo: 538, name: "Parth", age: 22, department: "IT", marks: 75 },
{ rollNo: 539, name: "Raj", age: 19, department: "ECE", marks: 89 },
{ rollNo: 540, name: "Sameer", age: 21, department: "ME", marks: 77 },
{ rollNo: 541, name: "Meera", age: 22, department: "CSE", marks: 74 },
{ rollNo: 542, name: "Tanvi", age: 20, department: "IT", marks: 63 },
{ rollNo: 543, name: "Aisha", age: 19, department: "ECE", marks: 62 },
{ rollNo: 544, name: "Kavya", age: 20, department: "ME", marks: 85 },
{ rollNo: 545, name: "Rhea", age: 20, department: "CSE", marks: 64 },
{ rollNo: 546, name: "Avni", age: 19, department: "IT", marks: 96 },
{ rollNo: 547, name: "Gauri", age: 20, department: "ECE", marks: 73 },
{ rollNo: 548, name: "Nitya", age: 21, department: "ME", marks: 85 },
{ rollNo: 549, name: "Sara", age: 21, department: "CSE", marks: 69 },
{ rollNo: 550, name: "Vanya", age: 20, department: "IT", marks: 68 },
{ rollNo: 551, name: "Lakshya", age: 19, department: "ECE", marks: 95 }}

> mongosh: Jaaduu



> MONGOSH

```
{ rollno: 551, name: 'Lakshya', age: 19, department: 'ECE', marks: 95 }
])
```

< {

acknowledged: true,

insertedIds: {

```
    '0': ObjectId('6ac331817a25504231dd0e2c'),
    '1': ObjectId('6ac331817a25504231dd0e2d'),
    '2': ObjectId('6ac331817a25504231dd0e2e'),
    '3': ObjectId('6ac331817a25504231dd0e2f'),
    '4': ObjectId('6ac331817a25504231dd0e30'),
    '5': ObjectId('6ac331817a25504231dd0e31'),
    '6': ObjectId('6ac331817a25504231dd0e32'),
    '7': ObjectId('6ac331817a25504231dd0e33'),
    '8': ObjectId('6ac331817a25504231dd0e34'),
    '9': ObjectId('6ac331817a25504231dd0e35'),
    '10': ObjectId('6ac331817a25504231dd0e36'),
    '11': ObjectId('6ac331817a25504231dd0e37'),
    '12': ObjectId('6ac331817a25504231dd0e38'),
    '13': ObjectId('6ac331817a25504231dd0e39'),
    '14': ObjectId('6ac331817a25504231dd0e3a'),
    '15': ObjectId('6ac331817a25504231dd0e3b'),
    '16': ObjectId('6ac331817a25504231dd0e3c'),
    '17': ObjectId('6ac331817a25504231dd0e3d'),
    '18': ObjectId('6ac331817a25504231dd0e3e'),
    '19': ObjectId('6ac331817a25504231dd0e3f'),
    '20': ObjectId('6ac331817a25504231dd0e40'),
    '21': ObjectId('6ac331817a25504231dd0e41'),
    '22': ObjectId('6ac331817a25504231dd0e42'),
    '23': ObjectId('6ac331817a25504231dd0e43'),
    '24': ObjectId('6ac331817a25504231dd0e44'),
    '25': ObjectId('6ac331817a25504231dd0e45'),
    '26': ObjectId('6ac331817a25504231dd0e46'),
    '27': ObjectId('6ac331817a25504231dd0e47'),
    '28': ObjectId('6ac331817a25504231dd0e48'),
    '29': ObjectId('6ac331817a25504231dd0e49'),
    '30': ObjectId('6ac331817a25504231dd0e4a'),
    '31': ObjectId('6ac331817a25504231dd0e4b'),
    '32': ObjectId('6ac331817a25504231dd0e4c'),
    '33': ObjectId('6ac331817a25504231dd0e4d'),
    '34': ObjectId('6ac331817a25504231dd0e4e'),
    '35': ObjectId('6ac331817a25504231dd0e4f'),
    '36': ObjectId('6ac331817a25504231dd0e50'),
```

3. Update OneUpdates

the first document that matches the given condition - here, the marks of roll number 501 are updated from 61 to 60.

Code:

```
db.students.updateOne(
  { rollNo: 501 },
  { $set: { marks: 60 } }
)
```

Output:

```
> db.students.updateOne(
  { rollNo: 501 },
  { $set: { marks: 60 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

ONE >

4. Update Many

Updates all documents that match the given condition - here, every student in the CSE department is given a new status field set to "Active".

Code:

```
db.students.updateMany(
  { department: "CSE" },
  { $set: { status: "Active" } }
```

)

Output:

```
> db.students.updateMany(
  { department: "CSE" },
  { $set: { status: "Active" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 25,
  modifiedCount: 25,
  upsertedCount: 0
}
```

ONE > |

5. Delete One

Deletes the first document that matches the given condition - here, the record of roll number 551 is removed.

Code:

```
db.students.deleteOne({ rollNo: 551 })
```

Output:

```
> db.students.deleteOne({ rollNo: 551 })
< {
  acknowledged: true,
  deletedCount: 1
}
ONE > |
```

6. Delete Many

Deletes all documents that match the given condition - here, every student with marks less than 40 is removed.

Code:

```
db.students.deleteMany({ marks: { $lt: 40 } })
```

Output:

```
> db.students.deleteMany({ marks: { $lt: 40 } })
< {
  acknowledged: true,
  deletedCount: 0
}
ONE >
```

7. Query Operations

A series of find() queries were run using MongoDB's comparison and logical operators to filter and retrieve student records.

Marks greater than 80:

Code: `db.students.find({ marks: { $gt: 80 } })`

Output:

```
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6aac16e3dc3bb82776d7ee98'),
  rollNo: 101,
  name: 'Rahul',
  age: 20,
  department: 'CSE',
  marks: 85,
  status: 'Active'
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee99'),
  rollNo: 102,
  name: 'Priya',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9b'),
  rollNo: 104,
  name: 'Neha',
  age: 22,
  department: 'CSE',
  marks: 88,
  status: 'Active'
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9d'),
  rollNo: 106,
  name: 'Ananya',
  age: 21,
  department: 'EE',
  marks: 94
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9f'),
  rollNo: 108,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 83
}
```

Code:

```
db.students.find({ marks: { $lt: 80 } })
```

Output:

```
>_ mongosh: Jaaduu +
>_MONGOSH
> db.students.find({ marks: { $lt: 80 } })
< {
  _id: ObjectId('6aac16e3dc3bb82776d7ee9a'),
  rollNo: 103,
  name: 'Amit',
  age: 19,
  department: 'ME',
  marks: 75
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9c'),
  rollNo: 105,
  name: 'Vikram',
  age: 20,
  department: 'Civil',
  marks: 67
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9e'),
  rollNo: 107,
  name: 'Rohit',
  age: 23,
  department: 'CSE',
  marks: 79,
  status: 'Active'
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7eea2'),
  rollNo: 111,
  name: 'Manish',
  age: 21,
  department: 'Civil',
  marks: 75
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7eea5'),
  rollNo: 114,
  name: 'Kavita',
  age: 21,
  department: 'EE',
  marks: 77
}
{
```

Code:

```
db.students.find({ marks: { $eq: 60 } })
```

Output:

```
> db.students.find({ marks: { $eq: 60 } })
< {
  _id: ObjectId('6ac3314a7a25504231dd0e2b'),
  rollNo: 501,
  name: 'Aarav',
  age: 18,
  department: 'CSE',
  marks: 60,
  status: 'Active'
}
{
  _id: ObjectId('6ac331817a25504231dd0e38'),
  rollNo: 514,
  name: 'Rudra',
  age: 20,
  department: 'IT',
  marks: 60
}
ONE >
```

Code:

```
db.students.find({ marks: { $gte: 80 } })
```

Output:

```
> db.students.find({ marks: { $gte: 80 } })
< {
  _id: ObjectId('6aac16e3dc3bb82776d7ee98'),
  rollNo: 101,
  name: 'Rahul',
  age: 20,
  department: 'CSE',
  marks: 85,
  status: 'Active'
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee99'),
  rollNo: 102,
  name: 'Priya',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9b'),
  rollNo: 104,
  name: 'Neha',
  age: 22,
  department: 'CSE',
  marks: 88,
  status: 'Active'
}
{
```

Code:

```
db.students.find({ marks: { $ne: 60 } })
```

This returns all 48 remaining students whose marks are not exactly 60.

Marks greater than 70 AND age less than 23:

```
> db.students.find({ marks: { $ne: 60 } })
< {
  _id: ObjectId('6aac16e3dc3bb82776d7ee98'),
  rollNo: 101,
  name: 'Rahul',
  age: 20,
  department: 'CSE',
  marks: 85,
  status: 'Active'
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee99'),
  rollNo: 102,
  name: 'Priya',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9a'),
  rollNo: 103,
  name: 'Amit',
  age: 19,
  department: 'ME',
  marks: 75
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9b'),
  rollNo: 104,
  name: 'Neha',
  age: 22,
  department: 'CSE',
  marks: 88,
  status: 'Active'
}
```

Code:

```
db.students.find({
  $and: [
    { marks: { $gt: 70 } },
    { age: { $lt: 23 } }
  ]
})
```

Output:

```
>_ mongosh: Jaaduu +
>_MONGOSH
> db.students.find({
  $and: [
    { marks: { $gt: 70 } },
    { age: { $lt: 23 } }
  ]
})
< [
  {
    _id: ObjectId('6aac16e3dc3bb82776d7ee98'),
    rollNo: 101,
    name: 'Rahul',
    age: 20,
    department: 'CSE',
    marks: 85,
    status: 'Active'
  },
  {
    _id: ObjectId('6aac16e3dc3bb82776d7ee99'),
    rollNo: 102,
    name: 'Priya',
    age: 21,
    department: 'ECE',
    marks: 90
  },
  {
    _id: ObjectId('6aac16e3dc3bb82776d7ee9a'),
    rollNo: 103,
    name: 'Amit',
    age: 19,
    department: 'ME',
    marks: 75
  },
  {
    _id: ObjectId('6aac16e3dc3bb82776d7ee9b'),
    rollNo: 104,
    name: 'Neha',
    age: 22,
    department: 'CSE',
    marks: 88,
    status: 'Active'
  }
]
```

Code:

```
db.students.find({ marks: { $not: { $gt: 80 } } })
```

Output:

```
> db.students.find({ marks: { $not: { $gt: 80 } } })
< {
  _id: ObjectId('6aac16e3dc3bb82776d7ee9a'),
  rollNo: 103,
  name: 'Amit',
  age: 19,
  department: 'ME',
  marks: 75
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9c'),
  rollNo: 105,
  name: 'Vikram',
  age: 20,
  department: 'Civil',
  marks: 67
}
{
  _id: ObjectId('6aac16e3dc3bb82776d7ee9e'),
  rollNo: 107,
  name: 'Rohit',
  age: 23,
  department: 'CSE',
  marks: 79,
  status: 'Active'
}
{
```